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# PSSM

*Rapport du  
Commandeur Data*

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# MONITOR

Creation des users Monitor qui prendrons le droit sudo par la suite.

```
sql@SQL:~$ su -  
Mot de passe :  
root@SQL:~# adduser monitor  
Nouveau mot de passe :  
Retapez le nouveau mot de passe :  
passwd : mot de passe mis à jour avec succès  
Modifier les informations associées à un utilisateur pour monitor  
Entrer la nouvelle valeur, ou appuyer sur ENTER pour la valeur par défaut  
  NOM []: monitor  
  Numéro de chambre []: 0  
  Téléphone professionnel []: 0  
  Téléphone personnel []: 0  
  Autre []: 0  
Is the information correct? [Y/n] y  
root@SQL:~# _
```

```
root@FTP:~# adduser monitor  
Nouveau mot de passe :  
Retapez le nouveau mot de passe :  
passwd : mot de passe mis à jour avec succès  
Modifier les informations associées à un utilisateur pour monitor  
Entrer la nouvelle valeur, ou appuyer sur ENTER pour la valeur par défaut  
  NOM []: monitor  
  Numéro de chambre []: 0  
  Téléphone professionnel []: 0  
  Téléphone personnel []: 0  
  Autre []: 0  
Is the information correct? [Y/n] y  
root@FTP:~# usermod -aG sudo monitor
```

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# MONITOR

Installation de SUDO.

```
root@WEB:~# apt update
Atteint : 1 http://deb.debian.org/debian trixie InRelease
Réception de : 2 http://deb.debian.org/debian trixie-updates InRelease [47,3 kB]
Réception de : 3 http://security.debian.org/debian-security trixie-security InRelease [43,4 kB]
Réception de : 4 http://security.debian.org/debian-security trixie-security/main Sources [230 kB]
Réception de : 5 http://security.debian.org/debian-security trixie-security/main amd64 Packages [1,1 MB]
Réception de : 6 http://security.debian.org/debian-security trixie-security/main Translation-en [1,1 MB]
742 ko réceptionnés en 1s (1 395 ko/s)
Tous les paquets sont à jour.
root@WEB:~# apt install -y sudo
Installation de :
  sudo

Sommaire :
  Mise à niveau de : 0. Installation de : 1 Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 2 086 kB
Espace nécessaire : 6 866 kB / 8 058 MB disponible

Réception de : 1 http://deb.debian.org/debian trixie/main amd64 sudo amd64 1.9.16p2-3+deb13u2
2 086 ko réceptionnés en 0s (9 650 ko/s)
Sélection du paquet sudo précédemment désélectionné.
(Lecture de la base de données... 36003 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../sudo_1.9.16p2-3+deb13u2_amd64.deb ...
Dépaquetage de sudo (1.9.16p2-3+deb13u2) ...
Paramétrage de sudo (1.9.16p2-3+deb13u2) ...
```

```
root@WEB:~# su - monitor
monitor@WEB:~$ sudo whoami
[sudo] Mot de passe de monitor :
root
```

# MONITOR

Installation de python3.

```
clientftp@ClientFTP:~$ sudo apt update && apt install -y python3 python3-pip python3-venv
[sudo] Mot de passe de clientftp :
clientftp n'est pas dans le fichier sudoers.
clientftp@ClientFTP:~$
```

```
Tous les paquets sont à jour.
python3 est déjà la version la plus récente (3.13.5-1).
python3 passé en « installé manuellement ».
```

Installation de :

python3-pip python3-venv

Installation de dépendances :

|                           |                           |                        |                  |
|---------------------------|---------------------------|------------------------|------------------|
| binutils                  | g++-14-x86-64-linux-gnu   | libc-dev-bin           | libgprofng0      |
| binutils-common           | g++-x86-64-linux-gnu      | libc6-dev              | libhwasan0       |
| binutils-x86-64-linux-gnu | gcc                       | libcc1-0               | libisl23         |
| build-essential           | gcc-14                    | libcrypt-dev           | libitm1          |
| cpp                       | gcc-14-x86-64-linux-gnu   | libctf-nobfd0          | libjs-jquery     |
| cpp-14                    | gcc-x86-64-linux-gnu      | libctf0                | libjs-sphinxdoc  |
| cpp-14-x86-64-linux-gnu   | javascript-common         | libdpkg-perl           | libjs-underscore |
| cpp-x86-64-linux-gnu      | libalgorithm-diff-perl    | libexpat1-dev          | liblsan0         |
| dpkg-dev                  | libalgorithm-diff-xs-perl | libfakeroot            | libmpc3          |
| fakeroot                  | libalgorithm-merge-perl   | libfile-fcntllock-perl | libmpfr6         |
| g++                       | libasan8                  | libgcc-14-dev          | libpython3-dev   |
| g++-14                    | libbinutils               | libgomp1               | libpython3.13    |

Paquets suggérés :

|                |                           |              |                 |                    |
|----------------|---------------------------|--------------|-----------------|--------------------|
| binutils-doc   | cpp-14-doc                | gcc-14-doc   | flex            | gdb-x86-64-linux-g |
| gprofng-gui    | debian-keyring            | gcc-multilib | bison           | apache2            |
| binutils-gold  | debian-tag2upload-keyring | autoconf     | gdb             | lighttpd           |
| cpp-doc        | g++-multilib              | automake     | gcc-doc         | httpd              |
| gcc-14-locales | g++-14-multilib           | libtool      | gcc-14-multilib | libc-devtools      |

Sommaire :

Mise à niveau de : 0. Installation de : 70Supprimé : 0. Non mis à jour : 0

Taille du téléchargement : 96,0 MB

Espace nécessaire : 364 MB / 8 051 MB disponible

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# MONITOR

Installation de Mariadb.

```
Précédemment des actions différentes (« triggers ») pour l'ice bin (L'ici le mariadb) ...
clientftp@ClientFTP:~$ sudo apt install -y mariadb-client (client)
-bash: erreur de syntaxe près du symbole inattendu « ( »
clientftp@ClientFTP:~$ sudo apt install -y mariadb-client
[sudo] Mot de passe de clientftp :
Installation de :
  mariadb-client

Installation de dépendances :
  libconfig-inifiles-perl  libdbi-perl  libmariadb3  libpcre2-posix3  mariadb-client-core
  libdbd-mariadb-perl      libgpm2  libncurses6  libterm-readkey-perl  mariadb-common

Paquets suggérés :
  libclone-perl  libmldbm-perl  libnet-daemon-perl  libsql-statement-perl  gpm

Sommaire :
  Mise à niveau de : 0. Installation de : 12Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 5 519 kB
Espace nécessaire : 86,0 MB / 7 668 MB disponible
```

```
clientftp@ClientFTP:~$ mysql --version
mysql from 11.8.6-MariaDB, client 15.2 for debian-linux-gnu (x86_64) using  EditLine wrapper
```

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# MONITOR

Installation de Mariadb.

```
clientftp@ClientFTP:~$ sudo apt install -y ftp
Installation de :
  ftp

Installation de dépendances :
  tnftp

Sommaire :
  Mise à niveau de : 0. Installation de : 2Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 163 kB
Espace nécessaire : 315 kB / 7 581 MB disponible
```

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# MONITOR

création un environnement virtuel Python (venv).

```
clientftp@ClientFTP:~$ sudo apt install -y ftp
Installation de :
  ftp

Installation de dépendances :
  tnftp

Sommaire :
  Mise à niveau de : 0. Installation de : 2Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 163 kB
Espace nécessaire : 315 kB / 7 581 MB disponible
```

```
clientftp@ClientFTP:~$ cd
clientftp@ClientFTP:~$ python3 -m venv venv
clientftp@ClientFTP:~$ ls
venv
```

```
clientftp@ClientFTP:~$ source venv/bin/activate
(venv) clientftp@ClientFTP:~$ _
```

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# MONITOR

ping de connexion internet et ping entre vms.

```
(venv) clientftp@ClientFTP:~$ ping 192.168.200.153
PING 192.168.200.153 (192.168.200.153) 56(84) bytes of data.
64 bytes from 192.168.200.153: icmp_seq=1 ttl=64 time=1.06 ms
64 bytes from 192.168.200.153: icmp_seq=2 ttl=64 time=0.384 ms
64 bytes from 192.168.200.153: icmp_seq=3 ttl=64 time=0.366 ms
^C
--- 192.168.200.153 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.366/0.603/1.060/0.322 ms
(venv) clientftp@ClientFTP:~$ ping 192.168.200.151
PING 192.168.200.151 (192.168.200.151) 56(84) bytes of data.
64 bytes from 192.168.200.151: icmp_seq=1 ttl=64 time=1.07 ms
64 bytes from 192.168.200.151: icmp_seq=2 ttl=64 time=0.354 ms
64 bytes from 192.168.200.151: icmp_seq=3 ttl=64 time=0.352 ms
^C
--- 192.168.200.151 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.352/0.591/1.067/0.336 ms
(venv) clientftp@ClientFTP:~$ ping 192.168.200.155
PING 192.168.200.155 (192.168.200.155) 56(84) bytes of data.
64 bytes from 192.168.200.155: icmp_seq=1 ttl=64 time=2.21 ms
64 bytes from 192.168.200.155: icmp_seq=2 ttl=64 time=0.341 ms
64 bytes from 192.168.200.155: icmp_seq=3 ttl=64 time=0.424 ms
64 bytes from 192.168.200.155: icmp_seq=4 ttl=64 time=0.563 ms
64 bytes from 192.168.200.155: icmp_seq=5 ttl=64 time=0.496 ms
64 bytes from 192.168.200.155: icmp_seq=6 ttl=64 time=0.341 ms
64 bytes from 192.168.200.155: icmp_seq=7 ttl=64 time=0.341 ms
64 bytes from 192.168.200.155: icmp_seq=8 ttl=64 time=0.341 ms
^C
--- 192.168.200.155 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.318/0.668/2.213/0.590 ms
```

```
(venv) clientftp@ClientFTP:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=3.87 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=3.43 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=3.74 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 3.431/3.680/3.874/0.184 ms
```

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---

# MONITOR

mysql pour python.

```
(venv) clientftp@ClientFTP:~$ mysql --version
mysql from 11.8.6-MariaDB, client 15.2 for debian-linux-gnu (x86_64) using EditLine wrapper

(venv) clientftp@ClientFTP:~$ pip install mysql-connector-python
Collecting mysql-connector-python
  Downloading mysql_connector_python-26.7.0-cp313-cp313-manylinux_2_28_x86_64.whl.metadata (11
Downloading mysql_connector_python-26.7.0-cp313-cp313-manylinux_2_28_x86_64.whl (21.7 MB)
  _____ 21.7/21.7 MB 8.8 MB/s eta 0:00:00
Installing collected packages: mysql-connector-python
Successfully installed mysql-connector-python-26.7.0
(venv) clientftp@ClientFTP:~$
```

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---

# MONITOR

clé SSH

```
(venv) clientftp@ClientFTP:~$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/clientftp/.ssh/id_ed25519):
Created directory '/home/clientftp/.ssh'.
Enter passphrase for "/home/clientftp/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/clientftp/.ssh/id_ed25519
Your public key has been saved in /home/clientftp/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:Z0/Lyv9EWS0/BU+S9IYqxnY2Vz27HCYsZc/FQ7Cpx/g clientftp@ClientFTP
The key's randomart image is:
+-- [ED25519 256] --+
|                 .o. |
|                .+=. |
|               *=+X |
|      .   B.=X* |
|     S B+*=+Bo |
|    = B+=+ = |
|    +E.+ |
|    . . . |
|    O.... |
+----- [SHA256] -----+
```

```
(venv) clientftp@ClientFTP:~$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/clientftp/.ssh/id_ed25519):
Created directory '/home/clientftp/.ssh'.
Enter passphrase for "/home/clientftp/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/clientftp/.ssh/id_ed25519
Your public key has been saved in /home/clientftp/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:Z0/Lyv9EWS0/BU+S9IYqxnY2Vz27HCYsZc/FQ7Cpx/g clientftp@ClientFTP
The key's randomart image is:
+-- [ED25519 256] --+
|                 .o. |
|                .+=. |
|               *=+X |
|      .   B.=X* |
|     S B+*=+Bo |
|    = B+=+ = |
|    +E.+ |
|    . . . |
|    O.... |
+----- [SHA256] -----+
(venv) clientftp@ClientFTP:~$ ssh-copy-id monitor@192.168.200.155
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/clientftp/.ssh/id_ed25519"
The authenticity of host '192.168.200.155 (192.168.200.155)' can't be established.
ED25519 key fingerprint is SHA256:7lSrxs3a+vuARfT+QTT9ME8qS0WvDwwf2VLSVBjy4Vg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to
monitor@192.168.200.155's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'monitor@192.168.200.155'"
and check to make sure that only the key(s) you wanted were added.
```

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# MONITOR

clé SSH all vms

```
(venv) clientftp@ClientFTP:~$ ssh monitor@192.168.200.155
Linux FTP 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64
```

```
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
```

```
Connection to 192.168.200.155 closed.
(venv) clientftp@ClientFTP:~$ ssh monitor@192.168.200.153
Linux WEB 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64
```

```
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
```

```
Connection to 192.168.200.151 closed.
(venv) clientftp@ClientFTP:~$ ssh monitor@192.168.200.151
Linux SQL 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64
```

```
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
```

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# MONITOR

## CONFIG SSH

```
# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to "no" here!
PasswordAuthentication no
#PermitEmptyPasswords no

# Change to "yes" to enable keyboard-interactive authentication.  Depending
# on the system's configuration, this may involve passwords, challenge-response,
# one-time passwords or some combination of these and other methods.
# Beware issues with some PAM modules and threads.
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no

# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing.  If this is enabled, PAM authentication will
# be allowed through the KbdInteractiveAuthentication and
# PasswordAuthentication.  Depending on your PAM configuration,
# PAM authentication via KbdInteractiveAuthentication may bypass
# the setting of "PermitRootLogin prohibit-password".
# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes

#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes

# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes

# no default banner path
#Banner none

# Allow client to pass locale and color environment variables
AcceptEnv LANG LC_* COLORTERM NO_COLOR

# override default of no subsystems
Subsystem sftp /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#    X11Forwarding no
#    AllowTcpForwarding no
#    PermitTTY no
#    ForceCommand cvs server
```

---

# MONITOR

```
monitor@WEB:~$ sudo systemctl restart sshd
monitor@WEB:~$ exit
déconnexion
Connection to 192.168.200.153 closed.
(venv) clientftp@ClientFTP:~$ ssh monitor@192.168.200.153
Linux WEB 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64
```

```
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Sep 16 14:28:29 2026 from 192.168.200.149
monitor@WEB:~$
```

```
# Change to "yes" to enable keyboard-interactive authentication. Depending on
# the system's configuration, this may involve passwords, challenge-response,
# one-time passwords or some combination of these and other methods.
# Beware issues with some PAM modules and threads.
KbdInteractiveAuthentication no
```

```
# Kerberos options
#KerberosAuthentication no
```

```
monitor@FTP:~$ sudo systemctl restart sshd
monitor@FTP:~$ exit
déconnexion
Connection to 192.168.200.155 closed.
(venv) clientftp@ClientFTP:~$ ssh monitor@192.168.200.155
Linux FTP 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64
```

```
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Sep 16 14:31:58 2026 from 192.168.200.149
monitor@FTP:~$ _
```

---

# SQL

```
monitor@SQL:~$ sudo apt update && sudo apt install -y mariadb-server
[sudo] Mot de passe de monitor :
Atteint : 1 http://deb.debian.org/debian trixie InRelease
Réception de : 2 http://deb.debian.org/debian trixie-updates InRelease [47,3 kB]
Réception de : 3 http://security.debian.org/debian-security trixie-security InRelease [43,4 kB]
Réception de : 4 http://security.debian.org/debian-security trixie-security/main Sources [229 kB]
Réception de : 5 http://security.debian.org/debian-security trixie-security/main amd64 Packages [10,5 kB]
Réception de : 6 http://security.debian.org/debian-security trixie-security/main Translation-en [1,2 kB]
741 ko réceptionnés en 1s (957 ko/s)
Lecture des listes de paquets... 26%
```

```
monitor@SQL:~$ sudo systemctl status mariadb
• mariadb.service - MariaDB 11.8.6 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-16 14:49:43 CEST; 2min 34s ago
 Invocation: df7b7ccf0a4b450c81ea1e51a0bc1724
    Docs: man:mariabdb(8)
          https://mariadb.com/kb/en/library/systemd/
 Main PID: 6086 (mariabdb)
   Status: "Taking your SQL requests now..."
    Tasks: 10 (limit: 14995)
  Memory: 123.4M (peak: 128.2M)
     CPU: 4.776s
  CGroup: /system.slice/mariadb.service
          └─6086 /usr/sbin/mariabdb
```

```
monitor@SQL:~$ sudo mysql_secure_installation
sudo: mysql_secure_installation : commande introuvable
```

# SQL

```
monitor@SQL:~$ sudo mariadb-secure-installation
```

```
NOTE: MariaDB is secure by default in Debian. Running this script will be
useless at best, and misleading at worst. This script will be removed
in a future MariaDB release in Debian. Please read the exact distribution
terms for each program are described in the individual files in
/usr/share/doc/mariadb-server/README.Debian.gz for details.
```

```
Enter root user password or leave blank:
```

```
Enter current password for root (enter for none):
```

```
OK, successfully used password, moving on...
```

```
Setting the root password or using the unix_socket ensures that you
can log into the MariaDB root user without the proper authorisation.
```

```
You already have your root account protected, so you can safely skip
```

```
Switch to unix_socket authentication [Y/n] y
```

```
Enabled successfully (or at least no errors was emitted)!
```

```
Reloading privilege tables..
```

```
... Success!
```

```
You already have your root account protected, so you can safely skip
```

```
Change the root password? [Y/n]
```

```
clientftp@ClientFTP:~$ ssh monitor@192.168.200.151
```

```
Linux SQL 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64
```

```
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
```

```
Last login: Wed Sep 16 14:48:17 2026 from 192.168.200.149
```

```
monitor@SQL:~$ sudo systemctl status mariadb
```

```
[sudo] Mot de passe de monitor :
```

```
• mariadb.service - MariaDB 11.8.6 database server
```

```
Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
```

```
Active: active (running) since Wed 2026-09-16 14:49:43 CEST; 4 days ago
```

```
Invocation: df7b7ccf0a4b450c81ea1e51a0bc1724
```

```
Docs: man:mariabdd(8)
```

```
https://mariadb.com/kb/en/library/systemd/
```

```
Main PID: 6086 (mariabdd)
```

```
Status: "Taking your SQL requests now..."
```

```
Tasks: 9 (limit: 14995)
```

```
Memory: 123.4M (peak: 128.2M)
```

```
CPU: 5.447s
```

```
CGroup: /system.slice/mariadb.service
```

```
└─6086 /usr/sbin/mariabdd
```

```
sept. 16 14:49:40 SQL mariabdd[6086]: 2026-09-16 14:49:40 0 [Note] Plugin 'FEEDBACK' is disabled.
```

```
sept. 16 14:49:40 SQL mariabdd[6086]: 2026-09-16 14:49:40 0 [Note] Plugin 'wsrep-provider' is disabled.
```

```
sept. 16 14:49:40 SQL mariabdd[6086]: 2026-09-16 14:49:40 0 [Note] InnoDB: Loading buffer pool(s) from /var/lib/my
```

```
sept. 16 14:49:40 SQL mariabdd[6086]: 2026-09-16 14:49:40 0 [Note] InnoDB: Buffer pool(s) load completed at 260916
```

```
sept. 16 14:49:42 SQL mariabdd[6086]: 2026-09-16 14:49:42 0 [Note] Server socket created on IP: '127.0.0.1', port:
```

```
sept. 16 14:49:42 SQL mariabdd[6086]: 2026-09-16 14:49:42 0 [Note] mariabdd: Event Scheduler: Loaded 0 events
```

```
sept. 16 14:49:42 SQL mariabdd[6086]: 2026-09-16 14:49:42 0 [Note] /usr/sbin/mariabdd: ready for connections.
```

```
sept. 16 14:49:42 SQL mariabdd[6086]: Version: '11.8.6-MariaDB-0+deb13u1 from Debian' socket: '/run/mysqld/mysqld
```

```
sept. 16 14:49:43 SQL systemd[1]: Started mariadb.service - MariaDB 11.8.6 database server.
```

```
sept. 16 14:49:44 SQL /etc/mysql/debian-start[6118]: Checking for insecure root accounts.
```

```
lines 1-24/24 (END)
```

---

---

# SQL

```
monitor@SQL:~$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 41
Server version: 11.8.6-MariaDB-0+deb13u1 from Debian -- Please help get to 10k stars at https://github.com/MariaDB
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE USER 'psmm_user'@'%' IDENTIFIED BY 'x';
Query OK, 0 rows affected (0,002 sec)

MariaDB [(none)]> CREATE DATABASE psmm_logs;
Query OK, 1 row affected (0,001 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON psmm_logs.* TO 'psmm_user'@'%';
Query OK, 0 rows affected (0,001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES
->
```

```
MariaDB [(none)]> FLUSH PRIVILEGES
-> ;
Query OK, 0 rows affected (0,001 sec)

MariaDB [(none)]>
```

# PYTHON

```
monitor@SQL:~$ exit
déconnexion
Connection to 192.168.200.151 closed.
clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u psmm_user -p psmm_logs
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 31
Server version: 11.8.6-MariaDB-0+deb13u1 from Debian -- Please help get to 10k stars at https://github.com/MariaDB

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [psmm_logs]> exit
Bye
clientftp@ClientFTP:~$ source venv/bin/activate
(venv) clientftp@ClientFTP:~$ pip install paramiko
Collecting paramiko
  Downloading paramiko-5.0.0-py3-none-any.whl.metadata (3.7 kB)
Collecting bcrypt>=3.2 (from paramiko)
  Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl.metadata (10 kB)
Collecting cryptography>=3.3 (from paramiko)
  Downloading cryptography-50.0.1-cp311-abi3-manylinux_2_34_x86_64.whl.metadata (4.3 kB)
Collecting invoke>=2.0 (from paramiko)
  Downloading invoke-3.0.3-py3-none-any.whl.metadata (3.2 kB)
Collecting pynacl>=1.5 (from paramiko)
  Downloading pynacl-1.6.2-cp38-abi3-manylinux_2_34_x86_64.whl.metadata (10.0 kB)
Collecting cffi>=2.0.0 (from cryptography>=3.3->paramiko)
  Downloading cffi-2.1.1-cp313-cp313-manylinux2014_x86_64_manylinux_2_17_x86_64.whl.metadata (2.5 kB)
Collecting pycparser (from cffi>=2.0.0->cryptography>=3.3->paramiko)
  Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
Downloading paramiko-5.0.0-py3-none-any.whl (208 kB)
Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl (278 kB)
Downloading cryptography-50.0.1-cp311-abi3-manylinux_2_34_x86_64.whl (4.7 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 4.7/4.7 MB 9.2 MB/s eta 0:00:00
Downloading cffi-2.1.1-cp313-cp313-manylinux2014_x86_64_manylinux_2_17_x86_64.whl (221 kB)
Downloading invoke-3.0.3-py3-none-any.whl (160 kB)
Downloading pynacl-1.6.2-cp38-abi3-manylinux_2_34_x86_64.whl (1.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.4/1.4 MB 11.2 MB/s eta 0:00:00
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
Installing collected packages: pycparser, invoke, bcrypt, cffi, pynacl, cryptography, paramiko
Successfully installed bcrypt-5.0.0 cffi-2.1.1 cryptography-50.0.1 invoke-3.0.3 paramiko-5.0.0 pycparser-3.0 pyr
```

```
(venv) clientftp@ClientFTP:~$ python3 ssh_login.py
Connexion réussie à 192.168.200.151
--- Résultat de la commande ---
Sys. de fichiers Taille Utilisé Dispo Uti% Monté sur
udev                947M      0  947M   0% /dev
tmpfs               194M    764K  193M   1% /run
/dev/sda1           9,2G    1,6G  7,2G  18% /
tmpfs               968M      0  968M   0% /dev/shm
tmpfs               1,0M      0   1,0M   0% /run/credentials/systemd-journald.service
tmpfs               5,0M      0   5,0M   0% /run/lock
tmpfs               968M      0  968M   0% /tmp
tmpfs               1,0M      0   1,0M   0% /run/credentials/getty@tty1.service
tmpfs              194M    8,0K  194M   1% /run/user/1000
tmpfs               1,0M      0   1,0M   0% /run/credentials/systemd-networkd.service
tmpfs              194M    8,0K  194M   1% /run/user/1001

(venv) clientftp@ClientFTP:~$
```

---

# PYTHON

```
import paramiko

hostname = "192.168.200.151"
port = 22
username = "monitor"
key_path = "/home/clientftp/.ssh/id_ed25519"
sudo_password = "x"

commande = "apt update"

client = paramiko.SSHClient()
client.set_missing_host_key_policy(paramiko.AutoAddPolicy())

try:
    client.connect(hostname=hostname, port=port, username=username, key_filename=key_path)
    print(f"Connexion réussie à {hostname}")

    commande_sudo = f"sudo -S -p ' ' {commande}"
    stdin, stdout, stderr = client.exec_command(commande_sudo)
    stdin.write(sudo_password + "\n")
    stdin.flush()

    sortie = stdout.read().decode()
    erreur = stderr.read().decode()

    print("--- Résultat ---")
    print(sortie)
    if erreur:
        print("--- Erreur ---")
        print(erreur)
except Exception as e:
    print(f"Erreur : {e}")
finally:
    client.close()
```

```
(venv) clientftp@ClientFTP:~$ python3 ssh_login_sudo.py
Connexion réussie à 192.168.200.151
--- Résultat ---
Atteint : 1 http://deb.debian.org/debian trixie InRelease
Réception de : 2 http://deb.debian.org/debian trixie-updates InRelease [47,3 kB]
Réception de : 3 http://security.debian.org/debian-security trixie-security InRelease [43,4 kB]
Réception de : 4 http://security.debian.org/debian-security trixie-security/main Sources [231 kB]
Réception de : 5 http://security.debian.org/debian-security trixie-security/main amd64 Packages [262 kB]
Réception de : 6 http://security.debian.org/debian-security trixie-security/main Translation-en [160 kB]
744 ko réceptionnés en 0s (2 976 ko/s)
Lecture des listes de paquets...
Construction de l'arbre des dépendances...
Lecture des informations d'état...
3 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les voir.

--- Erreur ---
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.
```

---

# PYTHON

```
GNU nano 8.4 ssh_mysql.py
import mysql.connector

hostname= "192.168.200.151"
user = "psmm_user"
password = "x"
database = "psmm_logs"

try:
    connexion = mysql.connector.connect(
        host=hostname,
        user=user,
        password=password,
        database=database
    )

    if connexion.is_connected():
        print(f"Connexion réussie à la base '{database}' sur {hostname}")
        info = connexion.get_server_info()
        print(f"Version du serveur MariaDB : {info}")

except mysql.connector.Error as e:
    print(f"Erreure de connexion : {e}")

finally:
    if 'connexion' in locals() and connexion.is_connected():
        connexion.close()
        print("Connexion fermée")
```

```
(venv) clientftp@ClientFTP:~$ python3 ssh_mysql.py
Connexion réussie à la base 'psmm_logs' sur 192.168.200.151
/home/clientftp/ssh_mysql.py:18: DeprecationWarning: Call to deprecated function get_server_info. Reason:
The property counterpart 'server_info' should be used instead.

    info = connexion.get_server_info()
Version du serveur MariaDB : 11.8.6-MariaDB-0+deb13u1 from Debian
Connexion fermée
(venv) clientftp@ClientFTP:~$
```

# PYTHON

```
GNU nano 8.4                                creer_table.sql
CREATE TABLE tentatives_sql (id INT AUTO_INCREMENT PRIMARY KEY, compte VARCHAR(100), date_heure DATETIME, ip VARCHAR(45));
```

```
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u psmm_user -p psmm_logs -e "DESCRIBE tentatives_sql;"
Enter password:
```

| Field      | Type         | Null | Key | Default | Extra          |
|------------|--------------|------|-----|---------|----------------|
| id         | int(11)      | NO   | PRI | NULL    | auto_increment |
| compte     | varchar(100) | YES  |     | NULL    |                |
| date_heure | datetime     | YES  |     | NULL    |                |
| ip         | varchar(45)  | YES  |     | NULL    |                |

```
(venv) clientftp@ClientFTP:~$ _
```

---

# PYTHON

```
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u psmm_user -p psmm_logs < creer_table.sql
Enter password:
(venv) clientftp@ClientFTP:~$ _
```

```
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u psmm_user -p psmm_logs -e "DESCRIBE tentatives_sql;"
Enter password:
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| id         | int(11)   | NO   | PRI | NULL    | auto_increment |
| compte    | varchar(100) | YES  |     | NULL    |              |
| date_heure | datetime  | YES  |     | NULL    |              |
| ip        | varchar(45) | YES  |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u faux1 -p psmm_logs
Enter password:
ERROR 1045 (28000): Access denied for user 'faux1'@'192.168.200.149' (using password: YES)
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u faux2 -p psmm_logs
Enter password:
ERROR 1045 (28000): Access denied for user 'faux2'@'192.168.200.149' (using password: YES)
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u faux3 -p psmm_logs
Enter password:
ERROR 1045 (28000): Access denied for user 'faux3'@'192.168.200.149' (using password: YES)
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u faux4 -p psmm_logs
Enter password:
ERROR 1045 (28000): Access denied for user 'faux4'@'192.168.200.149' (using password: YES)
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u faux5 -p psmm_logs
Enter password:
ERROR 1045 (28000): Access denied for user 'faux5'@'192.168.200.149' (using password: YES)
(venv) clientftp@ClientFTP:~$ _
```

# PYTHON

```
GNU nano 8.4 ssh_mysql_error.py
import paramiko
import re
import mysql.connector
from datetime import datetime

hostname = "192.168.200.151"
port= 22
ssh_user = "monitor"
key_path = "/home/clientftp/.ssh/id_ed25519"

dh_host = "192.168.200.151"
db_user = "psmm_user"
db_password = "x"
db_name = "psmm_logs"

client = paramiko.SSHClient()
client.set_missing_host_key_policy(paramiko.AutoAddPolicy())

try:
    client.connect(hostname=hostname, port=port, username=ssh_user, key_filename=key_path)

    commande = 'journalctl -u mariadb --since "1 hour ago" -- no-pager'
    stdin, stdout, stderr = client.exec_command(commande)
    logs = stdout.read().decode()

    pattern = r"(\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2}).*Access denied for user '([^']+)'@'([^']+)'

    tentatives = []
    for ligne in logs.splitlines():
        match = re.search(pattern, ligne)
        if match:
            date_heure_str, compte, ip = match.group()
            date_heure = datetime.strptime(date_heure_str, "%Y-%m-%d %H:%M:%S")

    print(f"{len(tentatives)} tentative(s) trouvée(s)")

    if tentatives:
        connexion = mysql.connector.connect(host=db_host, user=db_user, password=db_password, database=db_name)
        curseur = connexion.cursor()

        for compte, date_heure, ip in tentatives:
            curseur.execute(
                "INSERT INTO tentatives_sql (compte, date_heure, ip) VALUES (%s, %s, %s)",
                (compte, date_heure, ip)
            )

        connexion.commit()
        print(f"{curseur.rowcount if curseur.rowcount else len(tentatives)} ligne(s) insérée(s) dans la base")
        curseur.close()
        connexion.close()

except Exception as e:
    print(f"Erreur : {e}")

finally:
    client.close()
```

---

---

# FAUX1.2.3.4

```
monitor@SQL:~$ ls -la /var/log/mysql/
ls: impossible d'accéder à '/var/log/mysql/': Aucun fichier ou dossier de ce nom
monitor@SQL:~$ sudo journalctl -u mariadb --since "15 minute ago"
sept. 23 15:09:12 SQL mariadb[7532]: 2026-09-23 15:09:12 38 [Warning] Access denied for user 'faux1'@'192.168.200.149' (using password: YES)
sept. 23 15:09:26 SQL mariadb[7532]: 2026-09-23 15:09:26 39 [Warning] Access denied for user 'faux2'@'192.168.200.149' (using password: YES)
sept. 23 15:09:32 SQL mariadb[7532]: 2026-09-23 15:09:32 40 [Warning] Access denied for user 'faux3'@'192.168.200.149' (using password: YES)
sept. 23 15:09:40 SQL mariadb[7532]: 2026-09-23 15:09:40 41 [Warning] Access denied for user 'faux4'@'192.168.200.149' (using password: YES)
sept. 23 15:09:48 SQL mariadb[7532]: 2026-09-23 15:09:48 42 [Warning] Access denied for user 'faux5'@'192.168.200.149' (using password: YES)
monitor@SQL:~$ _
```

```
(venv) clientftp@ClientFTP:~$ python3 ssh_mysql_error.py
0 tentative(s) trouvée(s)
```

```
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u faux6 -p psmm_logs
Enter password:
ERROR 1045 (28000): Access denied for user 'faux6'@'192.168.200.149' (using password: YES)
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u faux7 -p psmm_logs
Enter password:
ERROR 1045 (28000): Access denied for user 'faux7'@'192.168.200.149' (using password: YES)
(venv) clientftp@ClientFTP:~$ python3 ssh_mysql_error.py
2 tentative(s) trouvée(s)
Erreur : name 'db_host' is not defined
```

---

---

# FAUX1.2.3.4

```
(venv) clientftp@ClientFTP:~$ python3 ssh_mysql_error.py
2 tentative(s) trouvée(s)
2 ligne(s) insérée(s) dans la base
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u psmm_user -p psmm_logs -e "SELECT * FROM tentatives_sql;"
Enter password:
+----+-----+-----+-----+
| id | compte | date_heure          | ip          |
+----+-----+-----+-----+
| 1  | faux6  | 2026-09-28 10:45:20 | 192.168.200.149 |
| 2  | faux7  | 2026-09-28 10:45:29 | 192.168.200.149 |
+----+-----+-----+-----+
```

```
(venv) clientftp@ClientFTP:~$ python3 ssh_mysql_error.py
2 tentative(s) trouvée(s)
2 ligne(s) insérée(s) dans la base
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u psmm_user -p psmm_logs -e "SELECT * FROM tentatives_sql;"
Enter password:
+----+-----+-----+-----+
| id | compte | date_heure          | ip          |
+----+-----+-----+-----+
| 1  | faux6  | 2026-09-28 10:45:20 | 192.168.200.149 |
| 2  | faux7  | 2026-09-28 10:45:29 | 192.168.200.149 |
| 3  | faux6  | 2026-09-28 10:45:20 | 192.168.200.149 |
| 4  | faux7  | 2026-09-28 10:45:29 | 192.168.200.149 |
+----+-----+-----+-----+
```

```
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u psmm_user -p psmm_logs -e "SELECT * FROM tentatives_sql;"
Enter password:
+----+-----+-----+-----+
| id | compte | date_heure          | ip          |
+----+-----+-----+-----+
| 1  | faux6  | 2026-09-28 10:45:20 | 192.168.200.149 |
| 2  | faux7  | 2026-09-28 10:45:29 | 192.168.200.149 |
+----+-----+-----+-----+
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u psmm_user -p psmm_logs -e "SELECT * FROM tentatives_sql;"
Enter password:
+----+-----+-----+-----+
| id | compte | date_heure          | ip          |
+----+-----+-----+-----+
| 1  | faux6  | 2026-09-28 10:45:20 | 192.168.200.149 |
| 2  | faux7  | 2026-09-28 10:45:29 | 192.168.200.149 |
+----+-----+-----+-----+
(venv) clientftp@ClientFTP:~$ _
```

---

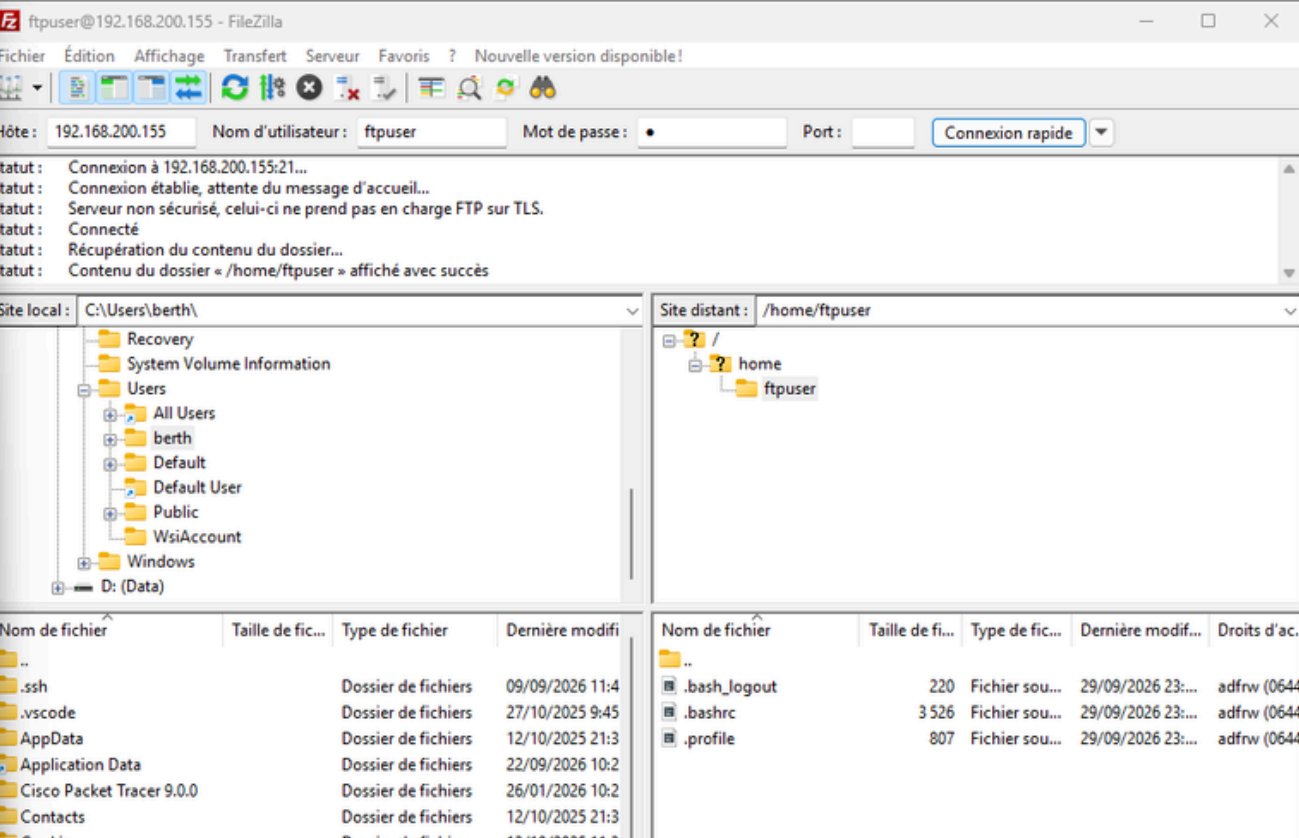
---

# FTP

```
monitor@FTP:~$ sudo apt update && sudo apt install -y proftpd
[sudo] Mot de passe de monitor :
Atteint : 1 http://deb.debian.org/debian trixie InRelease
Réception de : 2 http://deb.debian.org/debian trixie-updates InRelease [47,3 kB]
Réception de : 3 http://security.debian.org/debian-security trixie-security InRelease [43,4 kB]
Réception de : 4 http://security.debian.org/debian-security trixie-security/main Sources [233 k
Réception de : 5 http://security.debian.org/debian-security trixie-security/main amd64 Packages
Réception de : 6 http://security.debian.org/debian-security trixie-security/main Translation-en
749 ko réceptionnés en 1s (657 ko/s)
Lecture des listes de paquets... 13%
```

```
monitor@FTP:~$ sudo systemctl status proftpd
[sudo] Mot de passe de monitor :
• proftpd.service - ProFTPD FTP Server
   Loaded: loaded (/usr/lib/systemd/system/proftpd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-29 23:19:51 CEST; 22min ago
  Invocation: 70897b4104de4d9d885ec083cd48962b
     Docs: man:proftpd(8)
   Process: 5602 ExecStartPre=/usr/sbin/proftpd --configtest -c $CONFIG_FILE $OPTIONS (code=ex
   Process: 5604 ExecStart=/usr/sbin/proftpd -c $CONFIG_FILE $OPTIONS (code=exited, status=0/S
  Main PID: 5605 (proftpd)
     Tasks: 1 (limit: 1045)
    Memory: 2.3M (peak: 3.4M)
       CPU: 196ms
    CGroup: /system.slice/proftpd.service
            └─5605 "proftpd: (accepting connections)"

sept. 29 23:19:50 FTP systemd[1]: Starting proftpd.service - ProFTPD FTP Server...
sept. 29 23:19:51 FTP proftpd[5602]: Checking syntax of configuration file
sept. 29 23:19:51 FTP systemd[1]: Started proftpd.service - ProFTPD FTP Server.
monitor@FTP:~$
```



Connexion FTP non sécurisée

Ce serveur ne gère pas FTP sur TLS.  
Si vous continuez, votre mot de passe et vos fichiers seront envoyés en clair sur Internet.

Hôte : 192.168.200.155  
Port : 21

☐ Toujours autoriser FTP non sécurisé pour ce serveur.

OKAnnuler

---

# FTP

```
clientftp@ClientFTP:~$ ftp 192.168.200.155
Connected to 192.168.200.155.
220 ProFTPD Server (Debian) [::ffff:192.168.200.155]
Name (192.168.200.155:clientftp): faux1
421 Login timeout (300 seconds): closing control connection
ftp: Login failed
ftp> faux1
?Invalid command.
ftp> 192.168.200.155
?Invalid command.
ftp> exit
clientftp@ClientFTP:~$ ftp 192.168.200.155
Connected to 192.168.200.155.
220 ProFTPD Server (Debian) [::ffff:192.168.200.155]
Name (192.168.200.155:clientftp): faux1
331 Mot de passe requis pour faux1
Password:
530 Authentification incorrecte.
ftp: Login failed
ftp> faux2
?Invalid command.
ftp> exit
221 Au revoir.
clientftp@ClientFTP:~$ ftp 192.168.200.155
Connected to 192.168.200.155.
220 ProFTPD Server (Debian) [::ffff:192.168.200.155]
Name (192.168.200.155:clientftp): faux2
331 Mot de passe requis pour faux2
Password:
530 Authentification incorrecte.
ftp: Login failed
ftp> exit
421 Login timeout (300 seconds): closing control connection
clientftp@ClientFTP:~$ ftp 192.168.200.155
Connected to 192.168.200.155.
220 ProFTPD Server (Debian) [::ffff:192.168.200.155]
Name (192.168.200.155:clientftp): faux3
331 Mot de passe requis pour faux3
Password:
530 Authentification incorrecte.
ftp: Login failed
ftp> exit
221 Au revoir.
```

```
monitor@FTP:~$ sudo tail -20 /var/log/proftpd/proftpd.log
[sudo] Mot de passe de monitor :
2026-09-29 23:19:51,127 FTP proftpd[5605] FTP: ProFTPD 1.3.8c (maint) (built Wed Jul 22 2026 20:26:09 UTC)
2026-09-30 09:40:34,301 FTP proftpd[5699] FTP (192.168.200.1[192.168.200.1]): FTP no transfer timeout
2026-09-30 11:32:02,399 FTP proftpd[6581] FTP (192.168.200.149[192.168.200.149]): USER faux1: r
f:192.168.200.155:21
2026-09-30 11:33:45,280 FTP proftpd[6582] FTP (192.168.200.149[192.168.200.149]): USER faux2: r
f:192.168.200.155:21
2026-09-30 12:18:20,173 FTP proftpd[6639] FTP (192.168.200.149[192.168.200.149]): USER faux3: r
f:192.168.200.155:21
```

```
monitor@FTP:~$ sudo tail -20 /var/log/proftpd/proftpd.log
[sudo] Mot de passe de monitor :
2026-09-29 23:19:51,127 FTP proftpd[5605] FTP: ProFTPD 1.3.8c (maint) (built Wed Jul 22 2026 20:26:09 UTC)
2026-09-30 09:40:34,301 FTP proftpd[5699] FTP (192.168.200.1[192.168.200.1]): FTP no transfer timeout
2026-09-30 11:32:02,399 FTP proftpd[6581] FTP (192.168.200.149[192.168.200.149]): USER faux1: r
f:192.168.200.155:21
2026-09-30 11:33:45,280 FTP proftpd[6582] FTP (192.168.200.149[192.168.200.149]): USER faux2: r
f:192.168.200.155:21
2026-09-30 12:18:20,173 FTP proftpd[6639] FTP (192.168.200.149[192.168.200.149]): USER faux3: r
f:192.168.200.155:21
monitor@FTP:~$ sudo tail -5 /var/log/proftpd/proftpd.log - | fold -w 90
tail: option utilisée dans un contexte incorrect - 5
monitor@FTP:~$
```

# FTP

```
GNU nano 8.4 ssh_ftp_error.py
import paramiko
import re
import mysql.connector
from datetime import datetime

hostname = "192.168.200.155"
port = 22
ssh_user = "monitor"
key_path = "/home/clientftp/.ssh/id_ed25519"

db_host = "192.168.200.151"
db_user = "psmm_user"
db_password = "x"
db_name = "psmm_logs"

client = paramiko.SSHClient()
client.set_missing_host_key_policy(paramiko.AutoAddPolicy())

try:
    client.connect(hostname=hostname, port=port, username=ssh_user, key_filename=key_path)

    commande = "sudo cat /var/log/proftpd/proftpd.log"
    stdin, stdout, stderr = client.exec_command(commande)
    logs = stdout.read().decode()

    pattern = re.compile(
        r"(\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2}).*?USER (\S+): (.+?) from (\d+\.\d+\.\d+\.\d+)"
    )

    tentatives = []
    for match in pattern.finditer(logs):
        date_heure_str, compte, message, ip = match.groups()
        date_heure = datetime.strptime(date_heure_str, "%Y-%m-%d %H:%M:%S")
        tentatives.append((compte, date_heure, ip, message))

    print(f"{len(tentatives)} tentative(s) trouvee(s)")

except Exception as e:
    print(f"Erreur : {e}")

finally:
    client.close()
```

```
try:
    client.connect(hostname=hostname, port=port, username=ssh_user, key_filename=key_path)

    commande = "sudo cat /var/log/proftpd/proftpd.log"
    stdin, stdout, stderr = client.exec_command(commande)
    logs = stdout.read().decode()

    pattern = re.compile(
        r"(\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2}).*?USER (\S+): (.+?) from (\d+\.\d+\.\d+\.\d+)"
    )

    tentatives = []
    for match in pattern.finditer(logs):
        date_heure_str, compte, message, ip = match.groups()
        date_heure = datetime.strptime(date_heure_str, "%Y-%m-%d %H:%M:%S")
        tentatives.append((compte, date_heure, ip, message))

    print(f"{len(tentatives)} tentative(s) trouvee(s)")

    if tentatives:
        connexion = mysql.connector.connect(host=db_host, user=db_user, password=db_password, database=db_name)
        curseur = connexion.cursor()

        inserees = 0
        for compte, date_heure, ip, message in tentatives:
            curseur.execute(
                "INSERT IGNORE INTO tentatives_ftp (compte, date_heure, ip) VALUES (%s, %s, %s)",
                (compte, date_heure, ip)
            )
            inserees += curseur.rowcount

        connexion.commit()
        print(f"{inserees} nouvelle(s) ligne(s) inseree(s) sur {len(tentatives)} trouvee(s)")
        curseur.close()
        connexion.close()

except Exception as e:
    print(f"Erreur : {e}")

finally:
    client.close()
```

# FTP

```
Name (192.168.200.155:clientftp): monitor
331 Mot de passe requis pour monitor
Password:
530 Authentification incorrecte.
ftp: Login failed
ftp> exit
221 Au revoir.
(venv) clientftp@ClientFTP:~$ python3 ssh_ftp_error.py
6 tentative(s) trouvee(s)
Erreur : 1146 (42S02): Table 'psmm_logs.tentatives_ftp' doesn't exist
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u psmm_user -p psmm_logs -e "CREATE TABLE tentatives_ftp (id
INT AUTO_INCREMENT PRIMARY KEY, compte VARCHAR(100), date_heure DATETIME, ip VARCHAR(45), message VARCHAR(255), UNIQU
E KEY unique_tentative (compte, date_heure, ip));"
Enter password:
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u psmm_user -p psmm_logs -e "DESCRIBE tentatives_ftp;"
Enter password:
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int(11) | NO | PRI | NULL | auto_increment |
| compte | varchar(100) | YES | MUL | NULL |
| date_heure | datetime | YES | | NULL |
| ip    | varchar(45) | YES | | NULL |
| message | varchar(255) | YES | | NULL |
+-----+-----+-----+-----+-----+-----+

(venv) clientftp@ClientFTP:~$ python3 ssh_ftp_error.py
6 tentative(s) trouvee(s)
6 nouvelle(s) ligne(s) inseree(s) sur 6 trouvee(s)
(venv) clientftp@ClientFTP:~$ mysql -h 192.168.200.151 -u psmm_user -p psmm_logs -e "SELECT * FROM tentatives_ftp;"
Enter password:
+-----+-----+-----+-----+-----+-----+
| id | compte | date_heure | ip | message |
+-----+-----+-----+-----+-----+-----+
| 1 | faux1 | 2026-09-30 11:32:02 | 192.168.200.149 | no such user found from 192.168.200.149 [192.168.200.149] t
o ::ffff:192.168.200.155:21 |
| 2 | faux2 | 2026-09-30 11:33:45 | 192.168.200.149 | no such user found from 192.168.200.149 [192.168.200.149] t
o ::ffff:192.168.200.155:21 |
| 3 | faux3 | 2026-09-30 12:18:20 | 192.168.200.149 | no such user found from 192.168.200.149 [192.168.200.149] t
o ::ffff:192.168.200.155:21 |
| 4 | faux1000 | 2026-10-01 01:09:43 | 192.168.200.149 | no such user found from 192.168.200.149 [192.168.200.149] t
o ::ffff:192.168.200.155:21 |
| 5 | monitor | 2026-10-01 09:36:32 | 192.168.200.149 | (Login failed): Incorrect password |
| 6 | monitor | 2026-10-01 11:09:25 | 192.168.200.149 | (Login failed): Incorrect password |
+-----+-----+-----+-----+-----+-----+

(venv) clientftp@ClientFTP:~$
```

# WEB

```
monitor@WEB:~$ sudo apt install -y apache2 apache2-utils
Installation de :
  apache2  apache2-utils

Installation de dépendances :
  apache2-bin  libaprutil1-dbd-sqlite3  libcurl4t64  liblua5.4-0  libsasl2-2  libssh2-1t64
  apache2-data  libaprutil1-ldap  libldap-common  libnghttp3-9  libsasl2-modules  ssl-cert
  libapr1t64  libaprutil1t64  libldap2  librtmp1  libsasl2-modules-db

Paquets suggérés :
  apache2-doc  ufw  | libsasl2-modules-gssapi-heimdal  libsasl2-modules-sql
  apache2-suexec-pristine  www-browser  libsasl2-modules-ldap
  | apache2-suexec-custom  libsasl2-modules-gssapi-mit  libsasl2-modules-otp

Sommaire :
  Mise à niveau de : 0. Installation de : 19Supprimé : 0. Non mis à jour : 3
Taille du téléchargement : 3 542 kB
Espace nécessaire : 11,6 MB / 8 042 MB disponible

Réception de : 1 http://deb.debian.org/debian trixie/main amd64 libapr1t64 amd64 1.7.5-1 [104 kB]
Réception de : 2 http://deb.debian.org/debian trixie/main amd64 libaprutil1t64 amd64 1.6.3-3+deb13u1 [90,2
Réception de : 3 http://deb.debian.org/debian trixie/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-3+deb1
Réception de : 4 http://deb.debian.org/debian trixie/main amd64 libsasl2-modules-db amd64 2.1.28+dfsg1-9 [
Réception de : 5 http://deb.debian.org/debian trixie/main amd64 libsasl2-2 amd64 2.1.28+dfsg1-9 [57,5 kB]
Réception de : 6 http://deb.debian.org/debian trixie/main amd64 libldap2 amd64 2.6.10+dfsg-1 [194 kB]
Réception de : 7 http://deb.debian.org/debian trixie/main amd64 libaprutil1-ldap amd64 1.6.3-3+deb13u1 [12
Réception de : 8 http://deb.debian.org/debian trixie/main amd64 libnghttp3-9 amd64 1.8.0-1 [67,7 kB]
Réception de : 9 http://deb.debian.org/debian trixie/main amd64 librtmp1 amd64 2.4+20151223.git-fa8646d.1-2
Réception de : 10 http://deb.debian.org/debian trixie/main amd64 libssh2-1t64 amd64 1.11.1-1+deb13u2 [245
Réception de : 11 http://deb.debian.org/debian trixie/main amd64 libcurl4t64 amd64 8.14.1-2+deb13u5 [391 k
Réception de : 12 http://deb.debian.org/debian trixie/main amd64 liblua5.4-0 amd64 5.4.7-1+b2 [147 kB]
Réception de : 13 http://deb.debian.org/debian trixie/main amd64 apache2-bin amd64 2.4.68-1~deb13u1 [1 413
Réception de : 14 http://deb.debian.org/debian trixie/main amd64 apache2-data all 2.4.68-1~deb13u1 [160 kB]
Réception de : 15 http://deb.debian.org/debian trixie/main amd64 apache2-utils amd64 2.4.68-1~deb13u1 [220
Réception de : 16 http://deb.debian.org/debian trixie/main amd64 apache2 amd64 2.4.68-1~deb13u1 [228 kB]
```

```
monitor@WEB:~$ sudo htpasswd -bc /etc/apache2/
[sudo] Mot de passe de monitor :
Usage:
  htpasswd [-cimB25dpsDv] [-C cost] [-r rounds] passwordfile username
  htpasswd -b[cmB25dpsDv] [-C cost] [-r rounds] passwordfile username password

  htpasswd -n[imB25dps] [-C cost] [-r rounds] username
  htpasswd -nb[mB25dps] [-C cost] [-r rounds] username password

-c Create a new file.
-n Don't update file; display results on stdout.
-b Use the password from the command line rather than prompting for it.
-i Read password from stdin without verification (for script usage).
-m Force MD5 hashing of the password (default).
-2 Force SHA-256 hashing of the password (secure).
-5 Force SHA-512 hashing of the password (secure).
-B Force bcrypt hashing of the password (very secure).
-C Set the computing time used for the bcrypt algorithm
  (higher is more secure but slower, default: 5, valid: 4 to 17).
-r Set the number of rounds used for the SHA-256, SHA-512 algorithms
  (higher is more secure but slower, default: 5000).
-d Force CRYPT hashing of the password (8 chars max, insecure).
-s Force SHA-1 hashing of the password (insecure).
-p Do not hash the password (plaintext, insecure).
-D Delete the specified user.
-v Verify password for the specified user.
On other systems than Windows and NetWare the '-p' flag will probably not work.
The SHA-1 algorithm does not use a salt and is less secure than the MD5 algorithm.
```

# WEB

```
monitor@WEB:~$ sudo nano /etc/apache2/sites-enabled/000-default.conf
monitor@WEB:~$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Syntax OK
monitor@WEB:~$
```

```
# redirection URLs. In the context of virtual hosts, the ServerName
# specifies what hostname must appear in the request's Host: header to
# match this virtual host. For the default virtual host (this file) this
# value is not decisive as it is used as a last resort host regardless.
# However, you must set it for any further virtual host explicitly.
#ServerName www.example.com
```

```
ServerAdmin webmaster@localhost
DocumentRoot /var/www/html
```

```
# Available loglevels: trace8, ..., tracel, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn
```

```
ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined
```

```
# For most configuration files from conf-available/, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example the
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf
```

```
<Directory "/var/www/html">
    AuthType Basic
    AuthName "Zone protege"
    AuthUserFile /etc/apache2/.htpasswd
    Require valid-user
</Directory>
```

```
</VirtualHost>
```

```
monitor@WEB:~$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Syntax OK
```

```
monitor@WEB:~$ sudo systemctl restart apache2
```

```
monitor@WEB:~$ sudo systemctl status apache2 --no-pager
```

```
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-10-05 17:53:38 CEST; 5s ago
 Invocation: ab3bcb36e60449e8b1f9f724eedc6723
    Docs: https://httpd.apache.org/docs/2.4/
  Process: 7663 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
 Main PID: 7666 (apache2)
    Tasks: 55 (limit: 1045)
  Memory: 4.9M (peak: 5.1M)
     CPU: 85ms
   CGroup: /system.slice/apache2.service
           └─7666 /usr/sbin/apache2 -k start
             └─7667 /usr/sbin/apache2 -k start
               └─7669 /usr/sbin/apache2 -k start
```

```
oct. 05 17:53:38 WEB systemd[1]: Starting apache2.service - The Apache HTTP Server...
```

```
oct. 05 17:53:38 WEB apachectl[7665]: AH00558: apache2: Could not reliably determine the server's fully qualif... message
```

```
oct. 05 17:53:38 WEB systemd[1]: Started apache2.service - The Apache HTTP Server.
```

```
Hint: Some lines were ellipsized, use -l to show in full.
```

```
monitor@WEB:~$
```

# WEB

```
monitor@WEB:~$ sudo htpasswd -c /etc/apache2/.htpasswd admin
```

```
New password:
```

```
Re-type new password:
```

```
Adding password for user admin
```

```
monitor@WEB:~$
```

```
monitor@WEB:~$ sudo cat /etc/apache2/.htpasswd
```

```
admin:$apr1$X5z2gMtA$lg9cYq0zoEabjye33EWnb0
```

```
monitor@WEB:~$ curl http://localhost
```

```
-bash: curl : commande introuvable
```

```
Package curl est déjà installé. Les dépendances sont satisfaites.
```

```
monitor@WEB:~$ sudo apt install curl -y
```

```
Installation de :
```

```
curl
```

```
Sommaire :
```

```
Mise à niveau de : 0. Installation de : 1Supprimé : 0. Non mis à jour :
```

```
Taille du téléchargement : 270 kB
```

```
Espace nécessaire : 507 kB / 8 028 MB disponible
```

```
Prétraitement des actions différées (« triggers ») pour man-db (2.10.1-1) ...
```

```
monitor@WEB:~$ curl http://localhost
```

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
```

```
<html><head>
```

```
<title>401 Unauthorized</title>
```

```
</head><body>
```

```
<h1>Unauthorized</h1>
```

```
<p>This server could not verify that you  
are authorized to access the document  
requested. Either you supplied the wrong  
credentials (e.g., bad password), or your  
browser doesn't understand how to supply  
the credentials required.</p>
```

```
<hr>
```

```
<address>Apache/2.4.68 (Debian) Server at localhost Port 80</address>
```

```
</body></html>
```

# WEB

```
monitor@WEB:~$ curl -u toto:test http://localhost
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested. Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
<hr>
<address>Apache/2.4.68 (Debian) Server at localhost Port 80</address>
</body></html>
monitor@WEB:~$ curl -u admin:mauvaismotdepasse http://localhost
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested. Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
<hr>
<address>Apache/2.4.68 (Debian) Server at localhost Port 80</address>
</body></html>
monitor@WEB:~$ curl -u utilisateur:abc123 http://localhost
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested. Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
<hr>
<address>Apache/2.4.68 (Debian) Server at localhost Port 80</address>
</body></html>
monitor@WEB:~$
```

```
maintenement des actions déclenchées (ex: triggers /> pour mail de (2.15.1.1) ...
monitor@WEB:~$ curl http://localhost
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested. Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
<hr>
<address>Apache/2.4.68 (Debian) Server at localhost Port 80</address>
</body></html>
```

```
monitor@WEB:~$ sudo tail -30 /var/log/apache2/access.log
::1 - - [05/Oct/2026:18:05:19 +0200] "GET / HTTP/1.1" 401 706 "-" "curl/8.14.1"
::1 - - [05/Oct/2026:18:08:42 +0200] "GET / HTTP/1.1" 401 706 "-" "curl/8.14.1"
::1 - faux [05/Oct/2026:18:16:07 +0200] "GET / HTTP/1.1" 401 706 "-" "curl/8.14.1"
::1 - faux [05/Oct/2026:18:16:12 +0200] "GET / HTTP/1.1" 401 706 "-" "curl/8.14.1"
::1 - toto [05/Oct/2026:18:16:29 +0200] "GET / HTTP/1.1" 401 706 "-" "curl/8.14.1"
::1 - admin [05/Oct/2026:18:16:37 +0200] "GET / HTTP/1.1" 401 706 "-" "curl/8.14.1"
::1 - utilisateur [05/Oct/2026:18:16:41 +0200] "GET / HTTP/1.1" 401 706 "-" "curl/8.14.1"
monitor@WEB:~$ sudo tail -30 /var/log/apache2/error.log
[Mon Oct 05 17:10:24.071420 2026] [mpm_event:notice] [pid 7339:tid 7339] AH00489: Apache/2.4.68 (Debian) configured --
resuming normal operations
[Mon Oct 05 17:10:24.071673 2026] [core:notice] [pid 7339:tid 7339] AH00094: Command line: '/usr/sbin/apache2'
[Mon Oct 05 17:44:31.046085 2026] [mpm_event:notice] [pid 7339:tid 7339] AH00491: caught SIGTERM, shutting down
[Mon Oct 05 17:53:38.650929 2026] [mpm_event:notice] [pid 7666:tid 7666] AH00489: Apache/2.4.68 (Debian) configured --
resuming normal operations
[Mon Oct 05 17:53:38.651488 2026] [core:notice] [pid 7666:tid 7666] AH00094: Command line: '/usr/sbin/apache2'
[Mon Oct 05 18:16:07.605834 2026] [auth_basic:error] [pid 7667:tid 7673] [client ::1:45660] AH01618: user faux not foun
d: /
[Mon Oct 05 18:16:12.125335 2026] [auth_basic:error] [pid 7669:tid 7700] [client ::1:33738] AH01618: user faux not foun
d: /
[Mon Oct 05 18:16:29.539735 2026] [auth_basic:error] [pid 7669:tid 7702] [client ::1:45740] AH01618: user toto not foun
d: /
[Mon Oct 05 18:16:37.268485 2026] [auth_basic:error] [pid 7667:tid 7676] [client ::1:39114] AH01617: user admin: authen
tication failure for "/": Password Mismatch
[Mon Oct 05 18:16:41.853495 2026] [auth_basic:error] [pid 7669:tid 7704] [client ::1:57662] AH01618: user utilisateur n
ot found: /
```

```
GNU nano 8.4 ssh_web_error.py
import paramiko
import re
import mysql.connector
from datetime import datetime

# =====
# CONFIGURATION SERVEUR WEB
# =====

WEB_HOST = "192.168.200.153"
WEB_PORT = 22
WEB_USER = "monitor"
SSH_KEY = "/home/clientftp/.ssh/id_ed25519"

# =====
# CONFIGURATION MARIADB
# =====

DB_HOST = "192.168.200.151"
DB_USER = "psmm_user"
DB_PASSWORD = "x"
DB_NAME = "psmm_logs"

# =====
# CONNEXION SSH
# =====

ssh = paramiko.SSHClient()
ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())

try:

    print("Connexion au serveur WEB...")

    ssh.connect(
        hostname=WEB_HOST,
        port=WEB_PORT,
        username=WEB_USER,
        key_filename=SSH_KEY
    )

    print("Connexion SSH OK")

    print(f"[len(tentatives)] nouvelle(s) tentative(s) trouvee(s)")

    # =====
    # INSERTION EN BASE
    # =====

    for compte, date_heure, ip, message in tentatives:

        cursor.execute(
            """
            INSERT INTO tentatives_web
            (compte, date_heure, ip, message)
            VALUES (%s, %s, %s, %s)
            """
        )

    db.commit()

    print(
        f"[len(tentatives)] ligne(s) ajoutee(s) "
        "dans tentatives_web"
    )

    # =====
    # FERMETURE
    # =====

    cursor.close()
    db.close()

except Exception as e:

    print("ERREUR :")
    print(e)
```

```
# =====
# LECTURE DU LOG APACHE
# =====

command = "sudo -n cat /var/log/apache2/error.log"

stdin, stdout, stderr = ssh.exec_command(command)

logs = stdout.read().decode("utf-8", errors="replace")
erreur = stderr.read().decode("utf-8", errors="replace")

if erreur:
    print("Erreur sudo/SSH :")
    print(erreur)

if not logs:
    print("Aucune donnée récupérée dans error.log")
    exit()

# =====
# CONNEXION A MARIADB
# =====

db = mysql.connector.connect(
    host=DB_HOST,
    user=DB_USER,
    password=DB_PASSWORD,
    database=DB_NAME
)

cursor = db.cursor()

print("Connexion MariaDB OK")

# =====
# DERNIERE TENTATIVE DEJA ENREGISTREE
# =====

cursor.execute(
    """
    SELECT MAX(date_heure)
    FROM tentatives_web
    """

)

# =====
# FERMETURE
# =====

cursor.close()
db.close()

except Exception as e:

    print("ERREUR :")
    print(e)

finally:

    ssh.close()

    print("Connexion SSH fermee.")
```

```
GNU nano 8.4 ssh_web_error.py

# =====
# LECTURE DU LOG APACHE
# =====

command = "sudo -n cat /var/log/apache2/error.log"

stdin, stdout, stderr = ssh.exec_command(command)

logs = stdout.read().decode("utf-8", errors="replace")
erreur = stderr.read().decode("utf-8", errors="replace")

if erreur:
    print("Erreur sudo/SSH :")
    print(erreur)

if not logs:
    print("Aucune donnée récupérée dans error.log")
    exit()

# =====
# CONNEXION A MARIADB
# =====

db = mysql.connector.connect(
    host=DB_HOST,
    user=DB_USER,
    password=DB_PASSWORD,
    database=DB_NAME
)

cursor = db.cursor()

print("Connexion MariaDB OK")

# =====
# DERNIERE TENTATIVE DEJA ENREGISTREE
# =====

cursor.execute(
    """
    SELECT MAX(date_heure)
    FROM tentatives_web
    """

)
```

```
"""

resultat = cursor.fetchone()

derniere_date = resultat[0]

# =====
# REGEX DES ERREURS APACHE
# =====

pattern = re.compile(
    r"""
    \[
        (?P<date>
            [A-Z][a-z]{2}\s+
            [A-Z][a-z]{2}\s+
            \d{1,2}\s+
            \d{2}:\d{2}:\d{2}\.\d{6}\s+
            \d{4}
        )
    \]
    .*?
    \[client\s+
        (?P<ip>.+):\d+
    \]
    \s+
    (?P<code>AH01617|AH01618)
    :
    \s*user\s+
    (?P<compte>[^\s]+)
    :?
    (?P<reste>.*)
    """
    , re.VERBOSE
)

# =====
# EXTRACTION DES NOUVELLES TENTATIVES
# =====
```

```
# =====
# EXTRACTION DES NOUVELLES TENTATIVES
# =====

tentatives = []

for match in pattern.finditer(logs):

    date_str = match.group("date")
    ip = match.group("ip")
    code = match.group("code")
    compte = match.group("compte")
    reste = match.group("reste").strip()

    # Conversion de la date Apache vers datetime
    date_heure = datetime.strptime(
        date_str,
        "%a %b %d %H:%M:%S.%f %Y"
    )

    # Ignorer les anciennes lignes déjà enregistrées
    if derniere_date is not None:
        if date_heure <= derniere_date:
            continue

    # Reconstruction du message complet
    message = f"{code}: user {compte}"

    if reste:
        message += f" {reste}"

    # Limite de la colonne VARCHAR(255)
    message = message[:255]

    tentatives.append(
        (
            compte,
            date_heure,
            ip,
            message
        )
    )

)
```

# WEB

```
monitor@WEB:~$ curl -u nouveau: mauvais http://localhost
curl: (6) Could not resolve host: mauvais
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested. Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
<hr>
<address>Apache/2.4.68 (Debian) Server at localhost Port 80</address>
</body></html>
```

```
(venv) clientftp@ClientFTP:~$ python3 ssh_web_error.py
Connexion au serveur WEB...
Connexion SSH OK
Connexion MariaDB OK
3 nouvelle(s) tentative(s) trouvee(s)
3 ligne(s) ajoutee(s) dans tentatives_web
Connexion SSH fermee.
```

```
MariaDB [psmm_logs]> SELECT * FROM tentatives_web;
+-----+-----+-----+-----+-----+
--+
| id | compte | date_heure | ip | message |
+-----+-----+-----+-----+-----+
--+
| 1 | faux | 2026-10-05 18:16:07 | ::1 | AH01618: user faux not found: / |
| 2 | faux | 2026-10-05 18:16:12 | ::1 | AH01618: user faux not found: / |
| 3 | toto | 2026-10-05 18:16:29 | ::1 | AH01618: user toto not found: / |
| 4 | admin | 2026-10-05 18:16:37 | ::1 | AH01617: user admin: authentication failure for "/": Password Mismatch |
| 5 | utilisateur | 2026-10-05 18:16:41 | ::1 | AH01618: user utilisateur not found: / |
| 16 | utilisateur | 2026-10-05 18:16:41 | ::1 | AH01618: user utilisateur not found: / |
| 17 | utilisateur | 2026-10-05 18:16:41 | ::1 | AH01618: user utilisateur not found: / |
| 18 | nouveau | 2026-10-05 19:07:37 | ::1 | AH01618: user nouveau not found: / |
| 19 | nouveau | 2026-10-05 19:08:00 | ::1 | AH01618: user nouveau not found: / |
+-----+-----+-----+-----+-----+
--+
9 rows in set (0,002 sec)
```

# WEB

```
GNU nano 8.4 ssh_serveur_mail.py
import mysql.connector
import smtplib
from email.mime.text import MIMEText
from email.mime.multipart import MIMEMultipart
from datetime import datetime, timedelta

db_host = "192.168.200.151"
db_user = "psmm_user"
db_password = "x"
db_name = "psmm_logs"

expediteur = "ton.compte@gmail.com"
mdp_expediteur = "xxxx xxxx xxxx xxxx"
destinataire = "admin@exemple.com"

try:
    connexion = mysql.connector.connect(host=db_host, user=db_user, password=db_password, database=db_name)
    curseur = connexion.cursor()

    hier = (datetime.now() - timedelta(days=1)).date()

    tables = {
        "SQL (MariaDB)": "tentatives_sql",
        "FTP": "tentatives_ftp",
        "WEB": "tentatives_web",
    }

    corps = f"Rapport des tentatives de connexion echouees - {hier.strftime('%d/%m/%Y')}\n"
    corps += "=" * 60 + "\n\n"

    total = 0
    for nom_affiche, nom_table in tables.items():
        curseur.execute(
            f"SELECT compte, date_heure, ip FROM {nom_table} WHERE DATE(date_heure) = %s ORDER BY date_heure"
            (hier,)
        )
        lignes = curseur.fetchall()
        corps += f"--- {nom_affiche} ({len(lignes)} tentative(s)) ---\n"
        for compte, date_heure, ip in lignes:
            corps += f" {date_heure.strftime('%H:%M:%S')} | compte={compte} | ip={ip}\n"
        corps += "\n"
        total += len(lignes)

    corps += f"Total : {total} tentative(s) de connexion echouee(s) sur la journee.\n"

    curseur.close()
    connexion.close()

    message = MIMEMultipart()
    message["From"] = expediteur
    message["To"] = destinataire
    message["Subject"] = f"Rapport PSMM - {hier.strftime('%d/%m/%Y')} ({total} tentative(s))"
    message.attach(MIMEText(corps, "plain", "utf-8"))

    serveur = smtplib.SMTP("smtp.gmail.com", 587)
    serveur.starttls()
    serveur.login(expediteur, mdp_expediteur)
    serveur.sendmail(expediteur, destinataire, message.as_string())
    serveur.quit()

    print(f"Mail envoye avec succes a {destinataire} ({total} tentative(s) reportee(s))")

except Exception as e:
    print(f"Erreur : {e}")
```

## Mot de passe d'application généré

### Mot de passe d'application pour votre appareil

### Comment l'utiliser ?

Accédez aux paramètres de votre compte Google dans l'application ou l'appareil que vous essayez de configurer. Remplacez le mot de passe par celui de 16 caractères indiqué ci-dessus.

Tout comme votre mot de passe classique, ce mot de passe spécifique à une application permet d'accorder un accès complet à votre compte Google. Étant donné que vous n'avez pas besoin de le mémoriser, ne le notez nulle part ni ne le partagez avec personne.

Terminé

## Rapport PSMM - 05/10/2026 (12 tentative(s))

10:54 (il y a 1 minute)

ion echouees - 05/10/2026

=====

--- SQL (MariaDB) (1 tentative(s)) ---

10:53:59 | compte=testhier1 | ip=10.0.0.1

--- FTP (1 tentative(s)) ---

10:54:15 | compte=testhier2 | ip=10.0.0.2

--- WEB (10 tentative(s)) ---

10:54:21 | compte=testhier3 | ip=10.0.0.3

18:16:07 | compte=faux | ip=::1

18:16:12 | compte=faux | ip=::1

18:16:29 | compte=toto | ip=::1

18:16:37 | compte=admin | ip=::1

18:16:41 | compte=utilisateur | ip=::1

18:16:41 | compte=utilisateur | ip=::1

18:16:41 | compte=utilisateur | ip=::1

19:07:37 | compte=nouveau | ip=::1

19:08:00 | compte=nouveau | ip=::1

Total : 12 tentative(s) de connexion echouee(s) sur la journee.

```
MariaDB [psmm_logs]> INSERT INTO tentatives_sql (compte, date_heure, ip) VALUES ('testhier1', DATE_SUB(NOW(), INTERVAL 1 DAY), '10.0.0.1');
Query OK, 1 row affected (0,326 sec)

MariaDB [psmm_logs]> INSERT INTO tentatives_ftp (compte, date_heure, ip, message) VALUES ('testhier2', DATE_SUB(NOW(), INTERVAL 1 DAY), '10.0.0.2', 'no such user found');
Query OK, 1 row affected (0,060 sec)

MariaDB [psmm_logs]> INSERT INTO tentatives_web (compte, date_heure, ip, message) VALUES ('testhier3', DATE_SUB(NOW(), INTERVAL 1 DAY), '10.0.0.3', 'authentication failure');
Query OK, 1 row affected (0,002 sec)

MariaDB [psmm_logs]> exit
```

```
(venv) clientftp@ClientFTP:~$ python3 ssh_serveur_mail.py
Mail envoye avec succes a (12 tentative(s) reportee(s))
```

---

---

# MERCI !

---

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